

Industry Partner



Hackathon Partner



Project Name: Kairos

Problem Statement 8: AI for Industrial Knowledge Intelligence: Unified Asset & Operations Brain

Team Name: Deterium

College Name: SRM Institute of Science and Technology, Kattankulathur, Chennai, Tamil Nadu

Team Details:

S.No.	Name	Role
1	Arnav Bansal	Frontend Developer
2	Krishna Somani	Backend Developer
3	M Arshad	AI/ML Engineer

Executive Summary

Industrial operations run on knowledge that is scattered across disconnected systems and concentrated in a generation of engineers approaching retirement. Per the challenge brief, a typical large plant operates across 7 to 12 disconnected document systems, and an estimated 25 percent of India's experienced industrial engineers will retire within the next decade. Independently, the pattern is well documented: knowledge workers lose roughly a fifth of the workweek to searching for and gathering information, a finding established by McKinsey in 2012 and consistent with later studies through the 2020s.

Every prior solution, whether document management, enterprise search, retrieval-augmented generation, or knowledge graphs, treats knowledge as something to be retrieved. This fails the most dangerous gaps: the ones nobody knows to search for. **KAIROS inverts this.** It treats operational events such as work order creation, permit issuance, and shift handover as triggers, and proactively delivers the knowledge a person needs before they ask, governed by a temporal knowledge graph, an immutable evidence vault, and formal change-management integration so that what gets pushed is accurate enough to act on.

Business impact in this document is a transparent model built on cited external benchmarks, not invented deployment numbers, because KAIROS has not yet been deployed at a live facility. Prototype performance figures reported in Section 12 are measured results from our own pipeline runs, with sample sizes and scope stated explicitly.

1. The Problem

Per the challenge brief, professionals in asset-intensive industries lose a large share of working hours to searching for information and recreating documents that already exist. A typical large Indian plant operates across an estimated 7 to 12 disconnected document systems: P&IDs in one place, work orders in another, procedures in a third, and inspection records in a fourth. Fragmentation is cited as a contributing factor in unplanned downtime, and an estimated 25 percent of India's experienced industrial engineers will retire within the next decade, taking undocumented operational knowledge with them permanently.

The underlying pattern is independently corroborated. The McKinsey Global Institute (2012) found that knowledge workers spend roughly a fifth of the workweek, about 19 percent in the source, searching for and gathering information, a range consistent with later IDC and Interact studies through the 2020s. The IEEE Pulse of Engineering report (2019) found that over 60 percent of engineers rate knowledge loss on an employee's departure as very or extremely important, confirming that the retirement risk is a live, recognised concern rather than a hypothetical one.

This is a safety, compliance, and continuity problem that compounds every year a retiring cohort's tacit knowledge goes uncaptured.

2. Why Every Existing Solution Falls Short

Document management systems, enterprise search, RAG pipelines, knowledge graphs, and predictive maintenance platforms share one architectural assumption: knowledge must be retrieved. The model is always the same. Knowledge exists, a person has a need, they formulate a query, the system returns results. The best of these are excellent retrieval systems.

But retrieval cannot solve the most dangerous knowledge gaps: the ones nobody knows to query. A technician who does not know a pump has failed under this exact condition before does not know to search for its failure history. A planner who does not know a regulation changed last month does not know to search for the amendment. These are not retrieval failures. They are awareness failures, and no search engine, however sophisticated, can close a gap defined by the absence of a query.

3. What is KAIROS

KAIROS is the first industrial knowledge platform built around operational context awareness rather than query response. It continuously monitors a facility's operational pulse, interprets it against a governed temporal knowledge graph, and delivers the right knowledge to the right person at the moment it is needed, without being asked.

The inversion

KAIROS's job is not to answer questions. It ensures critical knowledge reaches people before they know they need it, treating events such as work order creation, permit-to-work issuance, shift handover, and alarm acknowledgment as triggers, not user actions.

The kairos moment

In Greek, kairos means the right, decisive moment, distinct from chronos, mere clock time. The unit of value is a contextually relevant operational briefing delivered when action is imminent.

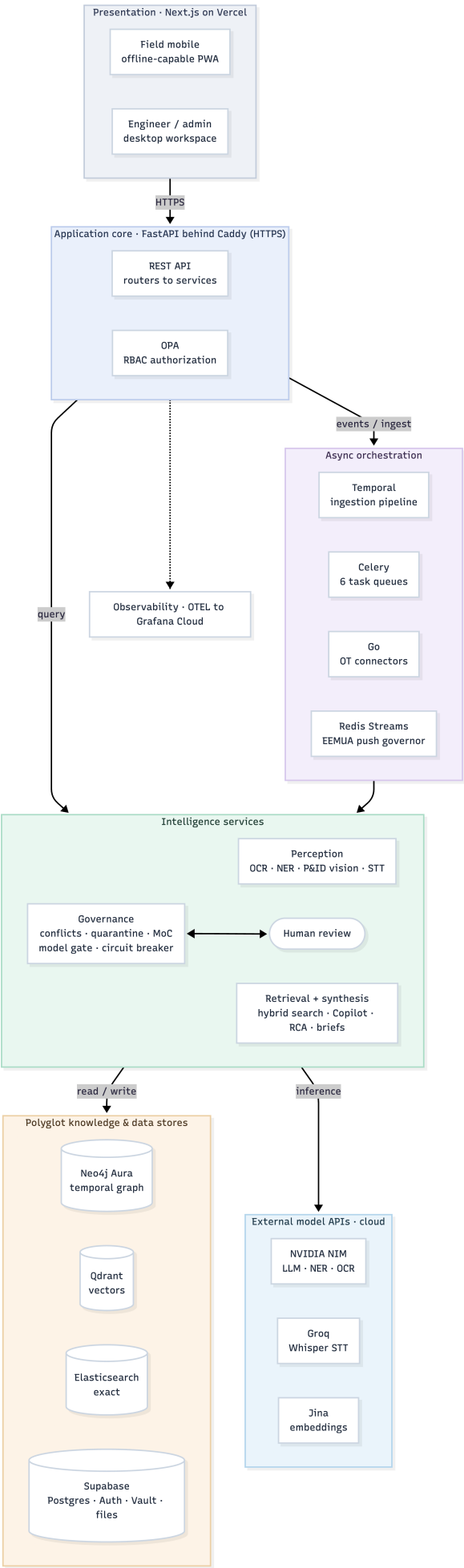
Why governance matters equally

Without proactive delivery, the result is a better document search tool. Without governed accuracy, proactive delivery becomes a confident hallucination machine wired into safety-critical workflows. KAIROS requires both.

Sector-agnostic: the architecture applies across oil and gas, power, pharmaceuticals, steel, mining, and any industry with complex assets and heterogeneous documentation. Regulatory standards differ by sector; the platform does not.

4. System Architecture

KAIROS is organised into thirteen layers across four zones. The reference architecture is shown in Figure 1; Section 5 summarises each layer's purpose.



5. The Thirteen-Layer Architecture

Each layer has a single, clearly bounded responsibility. This section summarises purpose; full visual detail is in Figure 1.

Zone 1, Foundation

- **Layer 0, Empirical Validation:** gates every model update against a rolling, human-verified validation corpus before production, class by class.
- **Layer 1, Deterministic Identity and MDM:** canonical asset identities imported from the enterprise ERP or EAM system; no AI-invented identities; human bootstrap for gaps.
- **Layer 2, Immutable Evidence Vault:** every source document is hashed, timestamped, and never overwritten.

Zone 2, Intelligence

- **Layer 3, Multimodal Perception Engine:** specialised OCR, named-entity recognition, and vision-based P&ID topology extraction, with mandatory engineer verification for safety-critical topology.
- **Layer 4, Temporal Reality Graph:** every fact is a time-bounded claim carrying a five-level authority hierarchy, provenance, and confidence score.
- **Layer 5, Zero-Copy OT Virtualisation:** live telemetry accessed via federated query, never bulk-ingested or duplicated.

Zone 3, Governance

- **Layer 6, Quarantine Layer:** a searchable, clearly labelled holding space for unverified field input.
- **Layer 7, Dual-Track Governance:** administrative conflicts route to lightweight review; safety-critical conflicts trigger formal Management of Change.

Zone 4, Delivery

- **Layer 8, Proactive Delivery Layer:** subscribes to operational events and delivers governed briefs before a person acts, under push-rate governance.
- **Layer 9, Structured Elicitation Engine:** short, graph-derived micro-interviews at the moment knowledge is freshest.
- **Layer 10, Outcome Attribution:** learns from telemetry and closeout data without corrupting itself with false causal inferences.
- **Layer 11, Reasoning and Synthesis:** assembles provenance-backed answers and refuses to guess on safety-critical low-confidence queries.
- **Layer 12, Phased Deployment and Interface:** earns trust through a staged rollout and delivers role-specific interfaces.

6. Document Corpus and Knowledge Sources

KAIROS ingests the full heterogeneous document landscape of a facility, in its actual state:

- **Engineering:** P&IDs, isolation diagrams, equipment datasheets, OEM manuals and bulletins
- **Operations:** work orders (SAP PM, Maximo, Infor EAM), maintenance history, shift logs, permits to work
- **Compliance:** operating and safety procedures, inspection reports, regulatory submissions, incident reports

- **Unstructured:** email archives, handwritten field notes, voice notes captured through the elicitation engine
- **Live systems:** OT historians such as OSIsoft PI, OPC-UA, and Honeywell Uniformance, accessed through zero-copy virtualisation, not bulk ingestion
- **Human expertise:** structured micro-interviews with experienced practitioners, including scheduled off-boarding sessions for retiring personnel

Every artifact enters through the same immutable evidence vault before any extraction occurs, giving the entire corpus a single auditable point of entry.

7. The Proactive Intelligence Engine

KAIROS subscribes to work order creation, permit-to-work issuance, shift handover, alarm acknowledgment, and other operational signals from source systems such as SAP PM and the site CMMS. A canonical event-normalisation step deduplicates, correlates, and reorders events arriving from multiple systems before anything reaches the trigger queue, preventing duplicate or out-of-sequence briefs.

Brief content by trigger

- **Work order brief:** the asset's full failure history, OEM guidance, and similar failures across the equipment class.
- **Permit brief:** exact isolation-point locations from drawing topology, device inspection status, and known physical deviations.
- **Shift handover brief:** open work orders, active alarms, and unresolved knowledge conflicts.

Push governance

Notification rates above roughly six events per operator per hour are established to degrade performance and cause reflexive dismissal. KAIROS enforces a hard per-user ceiling at that benchmark, backed by the EEMUA 191 and ISA-18.2 alarm-management standards, with a priority queue that never suppresses permit-triggered safety briefs. It requires thirty consecutive days of compliant push volume before proactive mode activates for any deployment.

8. Knowledge Governance and Trust Architecture

Proactive delivery without governed accuracy is a hallucination machine wired into safety-critical decisions. Every governance mechanism below exists to make proactive delivery safe, not as an afterthought.

- **The vault:** every source document is hashed, timestamped, and never overwritten, a legal and safety requirement given regulatory audit and investigation exposure.
- **The authority hierarchy:** every fact carries a five-level authority rank (regulatory requirement, engineering standard, OEM manual, site procedure, field observation). Conflicting facts are never averaged or resolved by recency; the system ranks by authority and escalates.
- **Blast-radius analysis:** when a source is superseded, every downstream fact and recommendation derived from it is automatically flagged for review, preventing silent propagation of outdated information.
- **Dual-track governance:** administrative conflicts get a lightweight review queue with a five-day service level; safety-critical conflicts trigger a formal Management of Change workflow with a 24-hour to five-day service level based on criticality. KAIROS auto-drafts the change request, the canonical graph is not updated until a signed resolution returns, and a warning banner displays on every affected query until then.

9. End-to-End Operational Flows

- **Flow A, Work order to proactive brief:** a work order opens for a critical asset. Within seconds, KAIROS queries the graph for failure history, OEM guidance, and similar failures across comparable assets. A synthesised, source-cited brief reaches the technician's phone before they leave the workshop.
- **Flow B, Permit issuance to safety brief:** an engineer issues a permit to work. KAIROS assembles the isolation boundary from drawing topology, the inspection status of every isolation device, and any flagged deviations. Both the issuing engineer and the shift lead must digitally sign off before work proceeds, with full evidence lineage logged.
- **Flow C, Document ingestion updates the graph:** a revised OEM bulletin enters the vault. The system detects a conflict with the existing pressure limit, runs blast-radius analysis across affected procedures, and auto-drafts a change request. Until signed, every affected query shows both values with a warning banner.
- **Flow D, Retiring engineer knowledge transfer:** the system identifies a retiring engineer as high knowledge density from her work order history and runs a scheduled series of short, graph-derived interviews on her areas of concentrated expertise. Her responses are reviewed and promoted to the canonical graph before her last day.

10. Intelligence and Response Features

- **Asset-centric truth model:** knowledge orbits equipment, not document folders.
- **Five-level authority hierarchy:** encodes real industrial governance into every graph edge.
- **Blast-radius analysis:** makes silent knowledge decay visible and actionable.
- **SPC-based adaptive circuit breaker:** per-asset-class thresholds instead of one static override rate.
- **Time-travel query:** accurate historical reconstruction for root cause investigations.
- **Zero-copy OT virtualisation:** telemetry context without historian-scale infrastructure complexity.
- **Proactive event-driven delivery:** the categorical differentiator from every retrieval-based competitor.
- **Permit-specific safety briefs with dual sign-off:** targets the single highest-leverage safety moment.
- **Context-derived micro-interview elicitation:** specific graph gaps produce specific, valuable answers.
- **Compliance intelligence as audit-preparation acceleration:** with an explicit false-negative bias.

11. User Experience and Interface Design

KAIROS delivers four role-specific interfaces rather than one generic dashboard.

- **Field interface:** a mobile app for real field conditions: single-handed operation, legible in direct sunlight, works offline with automatic sync, and supports voice input. The critical finding appears in the first two lines, and no menu navigation or query formulation is required to get an urgent answer.
- **Engineer and reliability desktop:** full graph visualisation, timeline view, conflict-resolution workflow, and audit-trail viewer for planners and reliability engineers working directly in the graph.
- **Quality and compliance cockpit:** a continuous compliance-gap dashboard, evidence-package assembly for audits, and non-conformance tracking for quality officers.
- **Management dashboard:** executive-level knowledge coverage, unresolved conflicts, and compliance posture for plant managers, without the operational detail underneath.

Design principles: zero training required for the field interface, zero extra steps beyond the existing workflow,

and every brief is actionable rather than merely informational.

12. Business Impact and ROI

The figures below are a transparent model built from cited external benchmarks, not measured deployment outcomes, because KAIROS has not yet been deployed at a live facility. Showing the arithmetic behind a modelled projection is more defensible than an unsupported point estimate.

Verified external benchmarks

- Knowledge workers lose roughly a fifth of the workweek, about 19 percent in the source, to searching for and gathering information (McKinsey Global Institute, 2012; consistent with later IDC and Interact studies).
- A searchable, governed knowledge record can reduce information-search time by as much as 35 percent, which the same report estimates returns approximately 6 percent of the workweek to other tasks (McKinsey Global Institute, 2012).
- Unplanned downtime costs Fortune Global 500 companies approximately 1.4 trillion US dollars annually, about 11 percent of total revenue, up from 8 percent (864 billion US dollars) in 2019 to 2020 (Siemens, True Cost of Downtime 2024).
- Over 60 percent of engineers rate knowledge loss on an employee's departure as very or extremely important (IEEE, Pulse of Engineering, 2019).

Modelled projection, arithmetic shown

The cleanest anchor is McKinsey's own downstream conclusion: reducing information-search time by up to 35 percent returns about 6 percent of the workweek to productive tasks. For a 45-hour week, that is roughly 2.7 hours per person per week. Across a fifty-person engineering and maintenance team, that is on the order of 135 hours per week returned, approximately three full-time equivalents of recovered capacity. KAIROS targets this recovery through proactive delivery rather than faster search.

Downtime framing

Set against the Siemens benchmark of 11 percent of revenue lost to unplanned downtime, even a small attributable reduction in mean time to repair, driven by delivering failure history before diagnosis begins rather than after, represents a meaningful line item at plant scale. We deliberately state no captured share, because no deployment data exists to support one.

Audit preparation, scoped honestly: KAIROS accelerates the desk-review portion of audit preparation. It does not replace mandatory field inspection or human sign-off. This document deliberately avoids stating a specific rupee figure or a downtime percentage attributed to KAIROS itself.

13. Validation and Evaluation Methodology

KAIROS validates against a rolling corpus built from real, human-verified operational data rather than a static curated dataset, so the benchmark stays representative as a deployment's document corpus evolves. Every model update must match or exceed the incumbent's precision and recall on the rolling corpus before promotion, class by class, so an aggregate improvement cannot mask a dangerous regression on a specific safety-relevant category.

Prototype validation results, measured

On our human-verified validation corpus, the entity-extraction model recorded an F1 of approximately 0.96 (precision 1.0, recall 0.92) in the reference run of 14 July 2026.

Scope and caveats, stated plainly: the ground-truth corpus for this run is small (13 entities), and the figures

are measured per entity type (asset tag, person, organisation), not per equipment class such as pump versus heat exchanger. Across repeated runs the observed range is F1 0.89 to 0.96 with recall near 0.92, reflecting a stochastic-model snapshot rather than a fixed constant.

Latency, measured: retrieval latency is approximately 1.4 seconds; end-to-end synthesis over 25 benchmark questions measured a median of 8.3 seconds and a 95th-percentile of 37 seconds using the 70B synthesis model.

Evaluation dimensions, aligned to the challenge's stated focus

- **Entity extraction accuracy:** benchmarked per entity type today, extending to per equipment class as the ground-truth corpus grows (measured, see above).
- **Query answer quality:** captured through the mandatory single-tap feedback (accurate, missing context, incorrect) that every synthesised output carries.
- **Knowledge graph linkage completeness:** tracked via a coverage dashboard that surfaces thin or expert-dependent areas explicitly.
- **Compliance gap detection:** held to an explicit false-negative bias, since an undetected real gap is a safety liability whereas a false positive is only a nuisance.
- **Time-to-answer versus traditional search:** a defined evaluation target, not yet measured against a keyword-search baseline; it becomes directly measurable during the retrieval-only phase of rollout.
- **P&ID topology extraction accuracy:** a defined target, functionally validated on a live drawing but not yet scored for precision and recall against an annotated ground truth.

14. Scalability: Day One to Enterprise Scale

- **Days 1 to 30, Retrieval mode:** MDM import establishes the asset skeleton and the vault begins ingesting documents. A single multilingual search interface across every system is available immediately, eliminating most document-hunting time even for a plant on 7 to 12 disconnected systems, without waiting for a complete graph.
- **Days 31 to 90, Assisted mode:** entity extraction and tag-alias resolution mature and the temporal graph populates with high-confidence facts. Around days 61 to 90, synthesised answers and the elicitation engine activate, gated on sixty days of established retrieval trust.
- **Around month 6, Governed proactive mode:** activates only after a thirty-day push-governance pilot confirms safe volumes. If the gate is not met, activation is deferred and configuration is revised rather than forced on a fixed date.
- **Multi-site scale:** a control plane manages enterprise-wide identity governance and cross-site failure-pattern detection, while each plant's data plane keeps its own graph and vault local, with cross-site sharing passing through the DPDP-aligned redaction pipeline first.

The primary engineering challenge, stated honestly: naive temporal-graph versioning can degrade catastrophically at enterprise scale. Performance engineering (composite indices, traversal-depth limits, precomputed hot views) must begin during the days 31 to 60 phase, not as late-stage hardening.

15. Built for India's Industrial Reality

English-only document processing loses a significant share of the operational knowledge captured in Hindi, Hinglish, and mixed-script maintenance forms, shift logs, and inspection records. KAIROS treats this as a core capability, not an edge case. Scanned and multilingual documents route through a cloud OCR model built for

mixed-script text, and entity extraction runs through a cloud NER model with a local offline fallback for air-gapped sites.

- **The honest problem:** multilingual and code-mixed named-entity recognition is genuinely harder than single-script English extraction, and no model handles it perfectly. The active-learning interface surfaces low-confidence extractions inline in normal search results for one-tap correction, building a facility-specific labelled dataset as a byproduct of ordinary use.
- **P&ID topology extraction:** standard OCR destroys the spatial relationships that carry the actual information in an engineering drawing. KAIROS uses vision understanding to reconstruct topology, with mandatory element-by-element engineer verification, since even commercial drawing parsers report only 70 to 80 percent automation rates.

DPDP Act 2023 alignment

India's Digital Personal Data Protection Act, 2023 is being implemented in phases: the Board was established effective 13 November 2025, the Consent Manager framework is effective 13 November 2026, and full obligations are effective 13 May 2027. Before any knowledge is promoted from a plant's local data plane to the central control plane, a redaction pipeline strips names, shift identifiers, and personal attributes, replacing them with role-generalised tokens, ahead of the final deadline rather than as a later retrofit.

Phase	Milestone	Effective
Phase 1	Data Protection Board established	13 November 2025 (complete)
Phase 2	Consent Manager framework	13 November 2026
Phase 3	Full obligations for all data fiduciaries	13 May 2027

16. Technical Stack

Category	Tools
Storage	Supabase Storage and PostgreSQL, Neo4j 5.20, Qdrant, Elasticsearch, Redis
AI and models	NVIDIA NIM (Llama 3.3 70B synthesis, Nemotron-OCR-v2, Ministral-14B NER), Ollama local fallback, Jina embeddings, Groq Whisper voice transcription
Backend	FastAPI, Go for high-throughput connectors, Celery, Temporal.io for durable workflows
Governance and security	Supabase Auth, Open Policy Agent, HashiCorp Vault, OpenTelemetry and Grafana
Frontend	Next.js 14 and React, Expo and React Native for mobile, shadcn/ui, Neovis.js for graph visualisation

Built fully for the demo

Document upload through OCR, NER, graph and vector search; hybrid retrieval and synthesis with citations; work order to proactive brief delivery; role-based auth; compliance gap detection.

Mocked for the demo, stated openly

The custom drawing parser (simplified topology used live), the alarm governor's underlying DCS feed, the cross-site control plane (single-site for the MVP), and the change-management webhook loop (manual approval stands in).

17. Edge Cases and Failure Handling

Scenario	Handling
MDM is incomplete or wrong	Human bootstrap required before AI linking; no fabricated identity is ever inserted.
Conflicting OEM manuals	Dual-track governance classifies severity; both versions preserved with authority levels shown.
Stale documents retrieved	Status is a first-class property; superseded documents never appear as current by default.
Poor scans below confidence threshold	Routed to human review; the original artifact is always available in the vault.
Synthesis layer unavailable	Retrieval continues independently; briefs fall back to raw results, never fully blinded.
OT historian unavailable	Document answers continue with an explicit notice; telemetry attribution pauses and resumes automatically.
Change request backlog grows	Warning banners persist; backlog surfaced to escalation authority; contested facts never silently treated as canonical.
Undocumented field modification found	Freezes dependent workflows until a site engineer confirms or a formal change request updates the record.
Expert refuses elicitation	Never mandatory; the gap is surfaced explicitly in the coverage dashboard, not hidden.
Cross-site knowledge conflict	Each site's governance stays independent; both engineering authorities review; discrepancy documented regardless of outcome.
False knowledge injected via elicitation	Enters quarantine only; cannot reach the canonical graph without human promotion.
Gradual model drift	Caught by the SPC circuit breaker in production and by the validation gate at every model update.
Adversarial data poisoning	Immutable vault enables forensic inspection; anomaly detection flags large parameter deviations; full audit trail enables rollback.

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